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Investigation of the melting of ionic liquid [emim][PF₆] confined inside carbon nanotubes using molecular dynamics simulations

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We have performed molecular dynamics (MD) simulations to study the thermodynamics, structure, and dynamical behavior of 1-ethyl-3-methylimidazolium hexafluorophosphate [emim][PF₆] during the melting process inside carbon nanotubes (CNTs) with different radii. Our structural results indicate the layering behavior for the IL inside the zigzag and armchair CNTs with the local maximums in the densities close to the CNT walls. The sharp and intense peaks are also observed at low temperature. However, as the temperature increases, the peaks related to the layer structure near the CNT wall become broader and less intense. It is also shown that the average number of hydrogen bonds decreases as the temperature increases. The internal energy also increases slightly with increasing temperature, followed by a dramatic jump. The temperature at this jump merely corresponds to the starting melting temperature, and another turning point in the energy curve corresponds to the entirely molten temperature, observed to be nearly 500 and 900 K, respectively. It is also shown that the diffusion coefficient increases as the temperature increases and a dramatic increase in the diffusion of the IL molecules occurs at the same temperature as the jump in total energy and thus marks the melting transition without ambiguity.

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1. Introduction

In recent years, interest in ionic liquids (ILs) has grown rapidly due to their desirable properties and various applications in many fields.^{1–6} Besides being extensively used in various homogeneous environments, ILs also exhibit exciting application potential in multiphase and confined systems, particularly in the areas of composite materials and electrochemistry.⁷ When confined inside nanometer-sized pores, ILs exhibit physical properties that are different from those observed in bulk systems.⁸ The understanding of the structure and phase behavior of confined ILs has attracted great attention due to their relevance in catalytic processes, lubricants, nanocomposites, fuels, and solar cells.^{8–19} For example, the melting points and the thermal decomposition temperature of several ILs have been reported to increase by more than 200 °C when confined inside multiwalled carbon nanotubes (MWCNTs).^{11,14} In contrast, when confined inside nanoporous silica,¹⁴ ILs can exhibit a similar mobility as the bulk phases, and their melting points decrease significantly.^{9,12} Also, Sha *et al.*¹⁵ reported a drastic phase transition in [Dmim][Cl] ionic liquid confined between two graphene walls. Zhang *et al.*¹⁶ observed an ordered ion arrangement of [bmim][PF₆] inside the channels of carbon nanotubes using

molecular dynamics (MD) simulation. Feng *et al.*¹⁷ also carried out MD simulations to investigate the solid–liquid transition of ionic liquid 1-ethyl-3-methyl imidazolium bromide ([emim]Br) by direct heating, hysteresis, void-nucleation, sandwich, and microcanonical ensemble approaches. They concluded that the void-nucleation and microcanonical ensemble methods are most favorable for predicting the solid–liquid transition.¹⁷ Dou *et al.*⁷ also investigated the melting process of 1-butyl-3-methylimidazolium hexafluorophosphate ([bmim][PF₆]) crystals confined in carbon nanotubes using MD simulation. They reported that the confined ions formed a “shell-chain” structure within the nanopore and this structure possessed long-range crystalline order at low temperature and initiated melting at approximately 500 K.⁷ Li *et al.*¹⁸ investigated [emim]Br ionic liquid which was confined in porous SiO₂ nanoparticles. They showed that the confinement in the nanopores leads to a compression of molecular size and an increase in the melting point. They also concluded that the pore size of SiO₂ is a key factor in tuning the anion cation distance and the melting point of the confined ionic liquids.¹⁸ In order to investigate the structure and dynamics of an ionic liquid confined at ambient temperature and pressure in hydroxylated amorphous silica nanopores, Ori *et al.*¹⁹ performed MD simulations in the isobaric–isothermal ensemble. They concluded that the most properties of the confined ionic liquid are a simple function of the surface-to-volume ratio of the host porous material.¹⁹

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It is very complicated to obtain a molecular-level picture of the behavior of ILs inside nanopores from experiments alone. Characterization techniques usually present an incomplete picture of the pore morphology of the porous material. Molecular simulations, on the other hand, allow us to work with a model porous material which is completely characterized. A reliable computational model can assist in the interpretation of the experimental results and provide accurate predictions of the behavior of the system.

The first goal of this research is to understand how thermodynamic, structural, and dynamical properties of an IL [emim][PF₆] confined inside the CNT are affected by the radius of the CNT during the melting process over a wide temperature range. Such fundamental understanding is crucial to optimize the performance of several alternative energy devices, such as electrochemical double layer capacitors (EDLCs) and dye-sensitized solar cells (DSSCs), as the molecular-level properties of confined ILs, affect the macroscopic performance of these devices.⁸

The second goal of this research is to understand how the thermodynamic, structural, and dynamical properties of the IL confined inside the CNT are affected by the chirality of the CNT during the melting process. We have considered the zigzag and armchair configurations. Such a fundamental investigation has not been performed before.

2. Simulation details

Constant volume and temperature (NVT) molecular dynamics simulations have performed using the DLPOLY 4.03 package. The simulations were carried out for the temperature from 300 to 1000 K. The Berendsen thermostat was applied to all of the species in the simulation box in order to maintain their temperature with a relaxation time of 0.1 fs. The equation motions have integrated using the verlet leapfrog algorithm with the integration time step of 0.5 fs. The SWCNTs were kept in fixed positions and were filled with varying amounts of the IL 1-ethyl-3-methylimidazolium hexafluorophosphate, [EMIM⁺][PF₆⁻]. The IL was modelled using the all-atom force field developed by Lopes *et al.*,^{20–22} which is based on the optimized potential for liquids simulation/all atom (OPLS-AA) of Jorgensen *et al.*²³

In order to prove the validity of force field, we carried out MD simulation of bulk [EMIM⁺][PF₆⁻] in the *NPT* ensemble. From these simulations we obtained a value of 1.541 g cm⁻³ for the bulk liquid density at 300 K. These simulated values are comparable to results from previous simulations,²⁴ and to reported experimental values (1.558 g cm⁻³).²⁵

For the intermolecular interactions between ionic liquid and SWCNT, we have used Lennard-Jones potential utilizing the geometric mean for ϵ and the arithmetic mean for σ . The columbic long range interactions were calculated using Ewalds' method with a precision of 10⁻⁶. All interatomic interactions between the atoms in the simulation box have calculated within the cutoff distance 15 Å. The periodic boundary conditions are applied only in the axial direction of the SWCNT. To avoid artificial influence from periodic images, IL-CNTs have kept in

the center of a simple orthorhombic box with vacuum on both sides separating it from the next periodic image in Z direction. The Z distance has kept large enough to avoid any interaction between ions and its periodic image. To analyze the effects of diameter and chirality of CNTs on the IL-CNTs interface, ten different SWCNTs have selected (such that all systems have a density of 1.541 g cm⁻³, which is similar to the model bulk density at 300 K (ref. 24)). The CNT systems chosen in the present studies include five armchair nanotubes and five zigzag nanotubes with diameters ranging from 1.5 to 3.5 nm. Simulations have run long enough to overcome the initial configurations. Here all productions runs have performed for 30 ns.

The average number of the hydrogen bonds between the confined IL molecules into the CNTs have been obtained using the calculation of the needed hydrogen bond length (2.67 Å) between the H atoms from the cations and the F atoms from the anions.^{7,25,26}

3. Results and discussion

3.1. Structural properties

We have presented the radial density profiles of the cations confined inside the zigzag ((20, 0), (26, 0), (45, 0)) and armchair ((15, 15), (26, 26)) CNTs with the different radii at different temperatures in Fig. 1 and 2 when the confined IL has a density similar to that of a bulk IL.

Fig. 1 and 2 show oscillations which indicates the layering behavior for the ions in the zigzag and armchair CNTs. It is also shown that the number of the layers increases from 1 to 4 as the radius of the nanotubes increases from 0.75 to 1.75 nm. According to Fig. 1 and 2, the local maximums in the mass density profiles of the confined IL in the densities close to the CNT walls which may be due to the stacking of the ions near the CNT wall surface. Our results presented in the Fig. 1 and 2 are in good agreement with the results of Shim and Kim²⁹ obtained for IL molecules inside single and double-walled CNTs. Singh *et al.*⁸

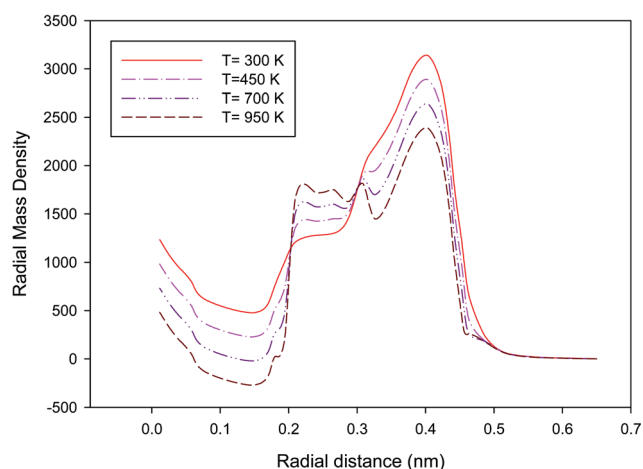


Fig. 1 The mass density profile of the confined cations into the smallest zigzag (20, 0) CNT along the radial direction at different temperatures (during the melting).

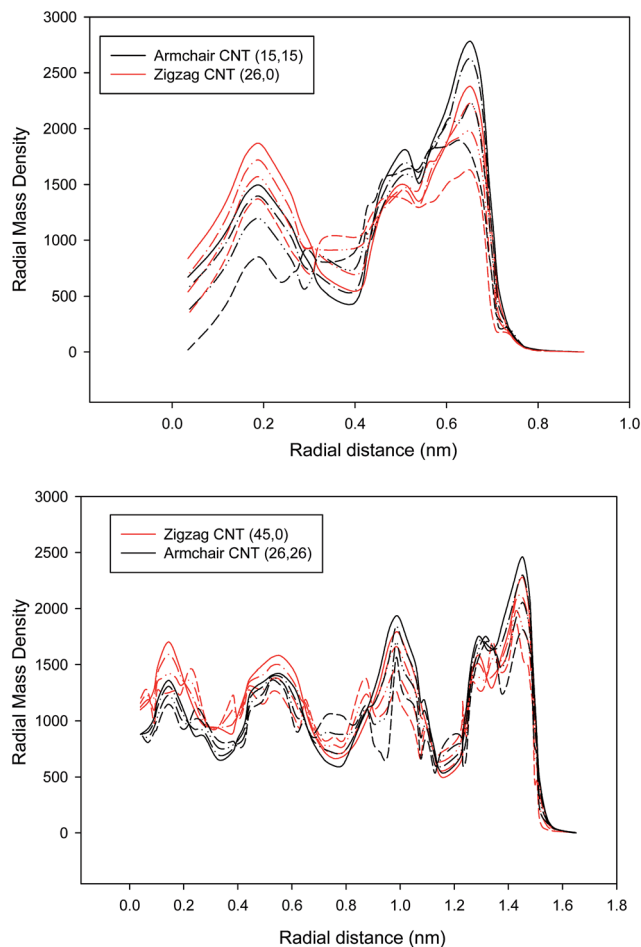


Fig. 2 The mass density profiles of the confined cations into the CNTs along the radial direction with the two different radii and chiralities at different temperatures (during the melting). The solid line is for $T = 300$ K, the dash-dot line is for $T = 450$ K, the dash-dot-dot line is for $T = 700$ K, and the dash line is for $T = 950$ K. The red and black colors represent the zigzag and armchair CNTs, respectively.

also concluded the similar results for different IL molecules inside MWCNTs.

The effect of temperature on the layer structure also manifests itself in changes of the radial number density profiles. As the peaks on the profiles in Fig. 1 and 2 indicate, the sharp and intense peaks are observed at low temperature (300 K). However, as the temperature increases, continuous changes may be observed. The peaks related to the layer structure near the CNT wall become broader and less intense. We have also presented the radial snapshots of the encapsulated ionic liquids within the nanotubes (26, 0) and (45, 0) before melting, at melting points, and after being molten entirely in Fig. 3. At low temperatures, the layering behavior can be observed for the ions in the CNTs. It is shown that the number of the layers increases as the radius of the nanotubes increases from 1 to 1.75 nm. It is also recognized that the layering behavior is greater for the smallest CNT which may be due to the greater interaction between the ions and the smaller wall than the bigger one. At higher temperatures, the

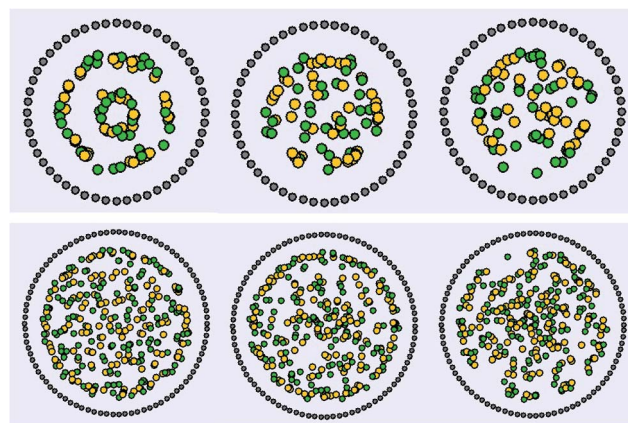


Fig. 3 The radial snapshots of the encapsulated ionic liquids within the nanotubes (26, 0) and (45, 0) before melting ($T = 300$ K), at melting points (Table 1) and after being molten entirely ($T = 1000$ K). Green and yellow balls represent center-of-mass locations of emim^+ and PF_6^- , respectively.

ions are more mobile and the ordered packing suffer serious damages. These results are in good agreement with the mass density profiles in Fig. 2.

For the first time, we have investigated the chirality effect in the radial density profiles of the IL molecules confined inside the CNTs in Fig. 1 and 2. At smallest radius ($R = 0.75$ nm), the only maximum in the mass density profile of the cations in the densities close to the zigzag CNT wall is higher than that of the armchair CNT which may be due to the greater interaction between the ions and the zigzag wall than the armchair wall. This can be due to the fact that inside a 0.75 nm CNT, the IL molecules are arranged in a single-file fashion, due to the small radius. Thus, the IL–IL interactions are restricted to the axial direction, allowing the IL–wall interactions to prevail. At larger CNTs, the inverse trend is found. In the other words, the maximums in the mass density profile of the confined ions in the densities close to the armchair CNT wall are higher than those of the zigzag CNT wall. It means that the changes in diameter of the CNT can lead to changes in the IL–wall interactions.

We have also investigated the axial density profiles of the cations confined inside the zigzag and armchair configurations for the smallest ($R = 0.75$ nm) and largest ($R = 1.75$ nm) CNTs in Fig. 4 and 5 at different temperatures, when the confined IL has a density similar to that of a bulk IL.

These figures show the local variations in the axial density profiles which exhibit regions of high and low densities. The similar observations have been obtained by Singh *et al.*⁸ for the axial density profiles of different IL molecules inside MWCNTs. Note that in case of the 0.75 nm CNT (Fig. 4), only a single array of the IL molecules is present inside the CNT and that gives rise to the oscillating density profiles. This result is in agreement with the result of Sadeghi and Parsafar²⁷ obtained for water molecules confined in CNTs with different sizes. The pronounced oscillatory behaviors of mass density profiles in Fig. 4 and 5 are clear signs that molecules occupy

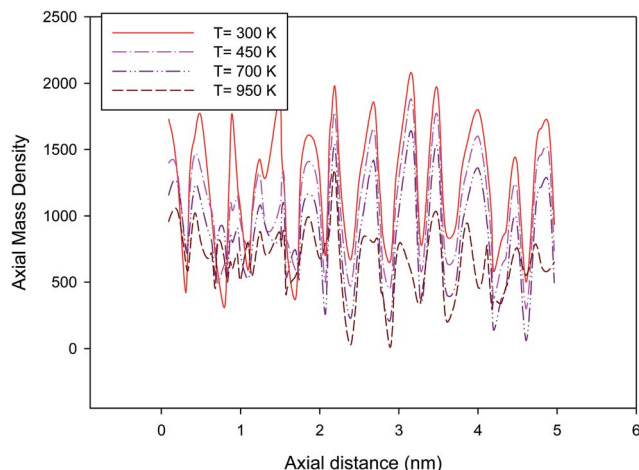


Fig. 4 The mass density profile of the confined cations into the smallest zigzag (20, 0) CNT along the axial direction at different temperatures (during the melting).

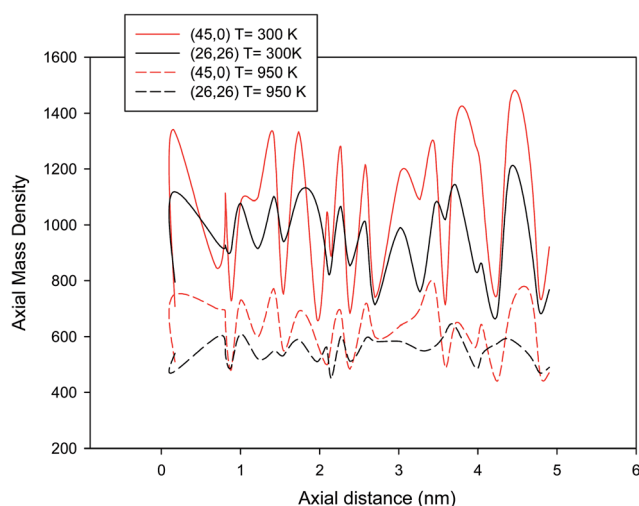


Fig. 5 The mass density profiles of the confined cations into the largest CNT along the axial direction at two different chiralities and temperatures.

well-defined locations inside the nanopore and thus form a more crystal-like structure than the liquid at low temperature ($T = 300$ K). For higher temperatures, the oscillatory character of the distribution becomes greatly reduced and its magnitude becomes nearly independent of the distance (Fig. 4 and 5). This suggests that internal solvation structure lacks crystal-like ordering. We have also observed in Fig. 5 that the local maximums in the axial density profiles are greater for the zigzag than the armchair configurations.

In order to investigate the structure of the IL molecules near the CNT wall, we have shown the average orientation of the cations as a function of the radial position for the CNTs with the different radii at $T = 300$ K when the confined IL has a density similar to that of a bulk IL in Fig. 6.

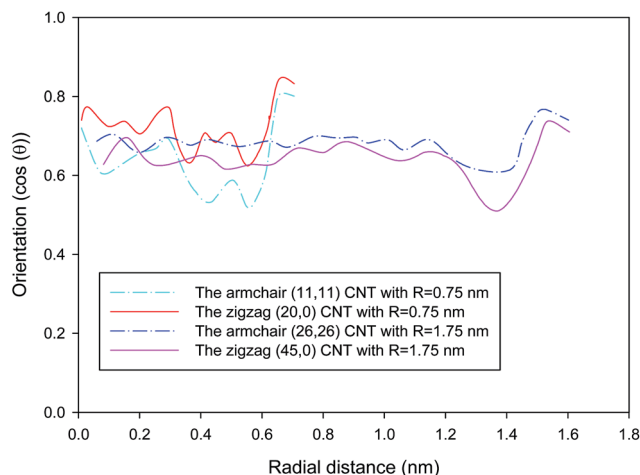


Fig. 6 The orientation distribution profile of the imidazolium ring of the cation as a function of radial position in the smallest and largest CNT with the different chiralities at 300 K.

In this figure, the orientation of the cations were measured by $\cos(\theta)$, where θ is the angle between the radial vector and the vector normal to the imidazolium ring. According to Fig. 6, the $\cos(\theta)$ has a relatively large value near the surfaces of the CNTs, indicating that the imidazolium ring of the cations tend to align parallel to the surface when close to the walls. In the smallest CNT, we have also observed that the $\cos(\theta)$ is larger for the cations into the zigzag CNT than the armchair configuration which means that the imidazolium ring of the cations in the zigzag configuration is more parallel to the wall surface than the armchair wall. This can be lead to the greater interaction between the IL and the smallest zigzag wall than the armchair wall. It is also shown that in the largest CNT, the imidazolium ring of the cations in the armchair configuration is more parallel to the wall surface than the zigzag wall. This can be lead to the inverse trend found in Fig. 2 in which we concluded that the IL has greater interaction with the armchair wall than the zigzag wall (in the largest CNT).

It is also shown in Fig. 6 that the average values of $\cos(\theta)$ drop before the last pick, suggesting a loss in orientational order with increasing distance from the CNT walls. The similar results obtained by Pinilla *et al.*²⁸ and Shim and Kim²⁹ for ILs confined inside slitlike pores and into single and double-walled CNTs, respectively. Singh *et al.*⁸ also concluded the same results for different IL molecules inside MWCNTs.

The average number of the hydrogen bonds have been also determined between the confined IL molecules into the CNTs with the smallest and largest CNTs and the both zigzag and armchair configurations in Fig. 7.

According to Fig. 7, the average number of the hydrogen bonds increase as the size of the CNTs decrease. It is also shown that the zigzag configurations are more suitable for the hydrogen bonds than the armchair CNT. In the other words, the most average number of the hydrogen bonds (6 to 7) forms in the smallest CNT ($R = 0.75$ nm) with the zigzag configuration at 300 K which may due to the presence of only a single array of the

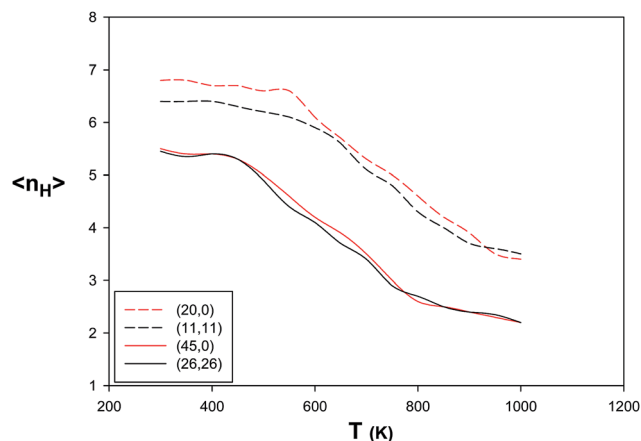


Fig. 7 The average number of hydrogen bonds between the cations and anions confined into the smallest and largest CNTs with the different chiralities at different temperatures.

IL molecules inside the CNT with the highest values of $\cos(\theta)$ near the surfaces of the CNT (Fig. 6), indicating that the imidazolium ring of the cations tend to align parallel to the surface which is more suitable to form the hydrogen bond. It can be also due to the greater interaction between the cations and anions in the first shell in the smallest zigzag CNT than the armchair configuration. Dou *et al.*⁷ also simulated the average number of the hydrogen bonds between 1-butyl-3-methylimidazolium hexafluorophosphate ([bmim][PF₆]) molecules inside a CNT with pore diameter of 1.94 nm. They found that the IL molecules form (the average) 6 hydrogen bonds at 300 K.

It is also shown in Fig. 7 that the average number of hydrogen bond decreases as the temperature increases. This can be due to the fact that as the amplitude of the ionic vibration increases with temperature, the average number of hydrogen bonds declines linearly. This decline might be directly responsible for the dramatic change in packing arrangement of ions inside the CNT. These changes in the molecular packing also affect the ionic diffusion behavior.⁷

3.2. Thermodynamic properties

The solid–liquid phase transition of confined liquid can be detected by monitoring the internal energy profiles.^{7,30–33} It is known that variation in the total energy of the system *versus* temperature shows different trends at different stages, such as before and during the melting processes and after being molten entirely.³⁴ In this study, we have calculated the internal energy of the IL confined inside the zigzag ((20, 0), (26, 0), (32, 0), (38, 0), (45, 0)) and armchair ((11, 11), (15, 15), (19, 19), (22, 22), (26, 26)) CNTs with the different radii at different temperatures, when the confined IL has a density similar to that of a bulk IL. The internal energy consists of kinetic energy and the cation–cation, cation–anion, and anion–anion intermolecular interactions and cation–walls and anion–walls interactions. We have also presented the internal energy of the IL inside the zigzag ((20, 0) and (45, 0))

and armchair ((11, 11) and (26, 26)) CNTs at different temperatures in Fig. 8. As the temperature rises, the energy value increases slightly, followed by a dramatic jump. The temperature at this jump merely corresponds to the starting melting temperature, and another turning point in the energy curve corresponds to the entirely molten temperature, observed to be nearly 500 and 900 K, respectively.

According to Fig. 8, the melting temperature (and its beginning and range) increases as the radius of the CNT decreases. This is in agreement with the previous studies of several ILs confined inside the MWCNTs.^{11,14} They found that the confinement effect increases the melting points and the thermal decomposition temperature of the ILs by more than 200 °C. It is also shown that the IL inside the smaller CNT has lower energy (and is more stable) than the bigger CNT which can be due to the greater interaction between the IL and the smaller CNT wall. In the other words, inside the smallest CNT ($R = 0.75$ nm), the IL molecules are arranged in a single-file fashion, due to the small radius. Thus, the IL–IL interactions are restricted to the axial direction, allowing the IL–wall interactions to prevail.

For the first time, we have investigated the chirality effect in the internal energy of the IL confined inside the CNTs during the melting process. According to Fig. 8, the IL inside the zigzag CNT has lower energy and is more stable than the armchair configuration which can be due to the greater interaction between the IL and the zigzag CNT than the armchair configuration. Therefore, the IL inside the zigzag CNT has higher melting temperature (and the melting beginning and its range) than the armchair CNT.

We have also presented the melting temperature ranges of the IL into the different CNTs in Table 1. According to this table, the melting temperatures (the beginnings and ranges) increase as the radius of the CNT decrease. It is also shown that the IL inside the zigzag CNTs have higher melting temperature than the armchair CNTs.

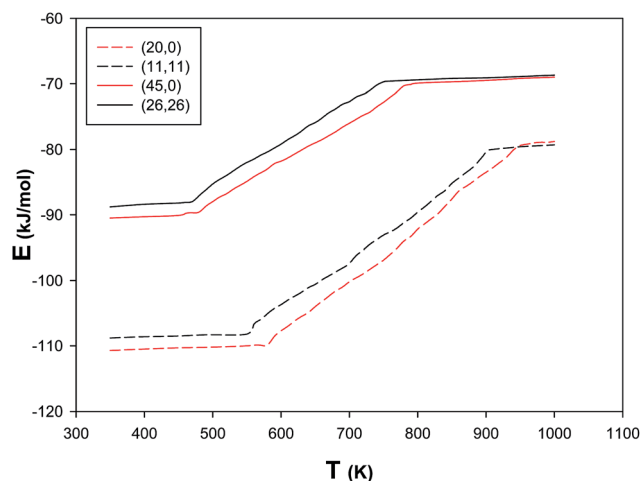
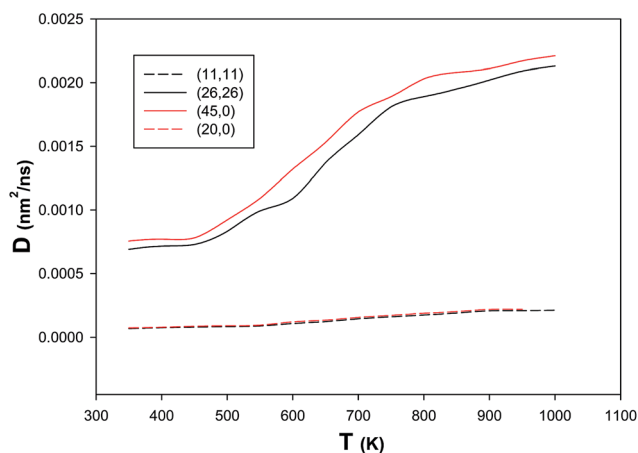


Fig. 8 The internal energy of the IL confined into the smallest and largest CNTs with the different chiralities at different temperatures (during the melting).

Table 1 The melting temperature ranges of the IL into the different CNTs

Zigzag CNTs				
(20, 0)	(26, 0)	(32, 0)	(38, 0)	(45, 0)
590–950 K	550–890 K	520–830 K	500–800 K	490–780 K
Armchair CNTs				
(11, 11)	(15, 15)	(19, 19)	(22, 22)	(26, 26)
560–900 K	530–850 K	500–800 K	490–780 K	480–750 K

**Fig. 9** The diffusion coefficient of the IL confined into the smallest and largest CNTs with the different chiralities at different temperatures.

3.3. Dynamic properties

We have also presented the results of the diffusion coefficient in the axial direction of the IL confined inside CNTs for the smallest and largest CNTs with the both zigzag and armchair configurations at different temperatures in Fig. 9.

Our diffusion results indicate that the IL molecules move faster into the larger CNT than the smaller one. The similar results obtained by Singh *et al.*⁸ for IL molecules inside MWCNTs. We have also found that the IL molecules have faster dynamics in the zigzag CNT than the armchair one which can be due to the more hydrogen bonds formed into the zigzag CNT than the armchair configuration. This is in agreement with the results of Taghavi *et al.*³⁵ They concluded that the hydrogen bond increases the diffusion rate of the confined molecules inside a nanotube. In the other words, the ordered collective motion of the IL molecules inside the zigzag CNT (due to the more hydrogen bonds) is faster than the motion into the armchair CNT.

It is also expected and shown in Fig. 9 that the diffusion coefficient increases as the temperature increases. Fig. 9 also shows a dramatic increase in the diffusion of the IL molecules occurs at the same temperature as the jump in total energy (Fig. 8) and thus marks clearly the melting transition.

4. Summary

We have performed the MD simulations to study the thermodynamics, structure, and dynamical behavior of the IL [emim]

[PF₆] during the melting process inside the CNTs with different radii ($R = 0.75, 1, 1.25$, and 1.5 nm) and chiralities (zigzag and armchair) at different temperatures when the confined IL has a density similar to that of a bulk IL.

Our results of the radial density profiles indicate the sharp and intense peaks at low temperature (300 K). However, as the temperature increases, continuous changes may be observed. The peaks related to the layer structure near the CNT wall become broader and less intense. Our results of the axial density profiles also show that the oscillatory character of the distribution becomes greatly reduced at higher temperatures. This suggests that internal solvation structure lacks crystal-like ordering. It is also shown that the average number of hydrogen bond decreases as the temperature increases. This can be due to the fact that as the amplitude of the ionic vibration increases with temperature, the average number of hydrogen bonds declines linearly.

Our results also indicate a dramatic jump in the energy values by increasing the temperature. The temperature at this jump merely corresponds to the starting melting temperature, and another turning point in the energy curve corresponds to the entirely molten temperature, observed to be nearly 500 and 900 K, respectively. It is also shown that the IL inside the zigzag CNT has lower energy and is more stable than the armchair configuration which can be due to the greater interaction between the IL and the zigzag CNT than the armchair configuration. Therefore, the IL inside the zigzag CNT has higher melting temperature than the armchair CNT. We have also found that the IL molecules have faster dynamics in the zigzag CNT than the armchair one which can be due to the more hydrogen bonds formed into the zigzag CNT than the armchair configuration.

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